



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS : X
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 03.10.2023

General Instructions:

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

[8 x 1 = 8]

1 i) If (2,2) are the coordinates of the centroid of a triangle whose vertices are (0, 4), (-2, 0) and (x, 2), then the value of x is:

- a) 2 b) 4 c) 6 d) 8

ii) If P(A) and P(A') are complementary events and P(A) = 0.15, then P(A') will be

- a) 0.35 b) 0.85 c) 0.3 d) cannot be determined

iii) The y-axis divides the line segment joining the points (-4, 5) and (3, -7) in the ratio:

- a) 2 : 7 b) 3 : 7 c) 4 : 3 d) 3 : 4

iv) If $\sin B = 0.8991$, then B is :

- a) $64^\circ 1'$ b) $64^\circ 2'$ c) $64^\circ 4'$ d) $64^\circ 6'$

v) If $\tan A = \cot B$, then :

- a) $A - B = 90^\circ$ b) $A = B = 90^\circ$ c) $A + B = 90^\circ$ d) $B - A = 90^\circ$

vi) If the length of the shadow of a tower is increasing, then the angle of elevation of the Sun is:

- a) Increasing b) decreasing c) not changed d) first increasing then decreasing

vii) If the point (a, 2a) lies on the line $y = 3x - 6$, then the value of a is :

- a) 1 b) 3 c) 6 d) 7

viii) The value of $\tan 30^\circ 26'$ is

- a) 0.5867 b) 0.5871 c) 0.5875 d) 0.5774

SECTION: B

2. Prove the following identities

a) $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$

b) $(\sin A + \cos A) (\tan A + \cot A) = \sec A + \operatorname{cosec} A$

[3 + 3 = 6]

3. a) Two coins are tossed once. Write the sample space. Find the probability of getting i) 2 heads ii) at least 1 tail.
- b) A die has 6 faces marked by the numbers -3, -2, -1, 0, 1 and 2. The die is thrown once. What is the probability of getting i) a positive integer ii) an integer greater than -3 iii) the smallest whole number. [3 + 3 = 6]
4. The angle of elevation of a jet plane from a point P on the ground is 60° . After a flight of 15 seconds the angle of elevation changes to 30° . If the jet plane is flying at a constant height of $1500\sqrt{3}$ m, find the speed of the jet plane. [4]
5. The vertices of a triangle ABC are A (3, 8), B (-1, 2) and C (6, -6). Find
a) Slope of BC
b) Equation of the line parallel to BC and passing through A [4]
6. Find the equation of line perpendicular to the line joining the points A (1, 2) and B (6, 7) and passing through the point which divides the line segment AB in the ratio 3:2 [4]
7. The top of a light house is 100m high. The angle of depression of two ships on opposite sides of its are 47° and 38° respectively. Find the distance between the ships to the nearest metre. [4]
8. ABCD is a parallelogram where A(x, y), B(5, 8), C (4, 7) and D(2, 4). Find the
i) coordinate of A ii) equation of diagonal BD. [4]